Emergent In-Group Bias: Sociality and Individuation in Robotic Swarms

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Abstract. The tendency to favour members of one's own group, also known as *in-group bias*, is ubiquitous in social mammals from rats to humans. This preference for in-group members (which often translates into aggression against out-group members) plays a major role in mammalian social life, and governs how individuals and groups perceive and behave toward others. Understanding how such social dynamics emerge, evolve, and impact behaviour is crucial for evolutionary theory and for the design of more mammal-like agents. Yet current swarm robots are typically extremely social and cooperative, and do not differentiate between individual robots. Here, we introduce two cognitive and social layers to a swarm engaged in a foraging task and a communication game: robot individuation (a partner-specific memory) and evolving sociality (a partner-specific tendency to interact based on previous interactions). We test whether social asymmetries such as in-group bias can spontaneously emerge from repeated local interactions without implicit group markers or pre-assigned identity tags, and study its strength and stability over varying conditions: cognitive (memory size and degree of decay), social (group size and innate sociality) and environmental (spread of resources). We show that implementing these two mammal-like features leads to the spontaneous emergence of in-group bias, which is robust across ecologies. These results shed light on the evolution of social asymmetries, and pave the way for modelling the complex social dynamics of mammals.

Keywords: Swarm robotics · In-group bias · Social cognition · Memory · Minimal Naming Game

1 Introduction

Cooperation is widely recognized as a key driver of evolutionary success [37], supporting diverse behaviours such as territorial defence [26], collective foraging [6], and the care of injured or vulnerable group members [28]. A rich body of research [19] has identified multiple mechanisms through which cooperative behaviours arise and are maintained, including kin selection [9], direct and indirect reciprocity [35], network reciprocity, and group selection [33]. In particular, models of group selection highlight that individuals often behave differently toward

members of their own group compared to outsiders, yielding intra-group benefits while simultaneously promoting inter-group differentiation and competition [23]. The expression and stability of cooperation are further shaped by ecological conditions [27] such as resource availability, spatial structure, and environmental risk, as well as by the interaction patterns and social mechanisms that govern populations [38].

In-group bias, namely, the tendency to favour and preferentially cooperate with individuals who are perceived as members of one's own group, has been documented across a wide range of species [15], including social insects [10], mammals [14], non-human primates [11], and humans. Although ubiquitous in the animal kingdom, the expression of this bias differs profoundly across taxonomic groups. In social insects such as ants, which constitute one of the primary biological inspirations for swarm robotics models [2], in-group favouritism is largely innate and genetically determined: colony-specific chemical cues allow individuals to identify nestmates [39]. In contrast, in mammals [32] and certainly in primates [36], in-group bias is flexible and acquired, and arises through learning, social experience, and the accumulation of partner-specific histories. These differences raise key evolutionary questions: to what extent is in-group bias an adaptive feature? How, and under which ecological conditions, does it emerge?

In-group bias has been investigated through a wide range of approaches, from naturalistic observations to mathematical models and agent-based simulations (see review in [15]). Although these models can provide valuable insights (e.g., on how the spread of resources impacts food sharing [24]), they are not without problems. The first issue is that many of these models rely on strong assumptions with respect to the target behaviour: most notably, individuals are typically a priori assigned to groups, for example through arbitrary tags [30]. These make group membership a pre-defined input of the system rather than an emergent outcome, and deviate from learned, mammal-like social dynamics. Second, most models rely on abstract agents that lack embodiment, spatial constraints, or physical interaction, even though these factors are known to strongly influence social behaviour and cooperation [21, 20, 12]. Swarm robotics models are thus particularly well positioned to address these gaps, and have already been used to model complex collective behaviours such as social insects foraging [25, 31, 4] and to provide insights into language evolution [34, 16, 3]. Here, we propose swarm robotics as the ideal framework for modelling this complex social layer, and to test whether social asymmetries such as in-group bias can emerge spontaneously, without any implicit group markers or pre-assigned identity tags.

As we are interested in modelling mammal-like social behaviour, an important first step is to create more mammal-like robots, with more complex behavioural and cognitive features. Specifically, swarm robotics typically assumes that robots are insect-like with relatively limited cognitive capabilities, such as minimal memory or restricted local sensors. However, a key difference between insect and mammal social systems is the role of partner-specific memory, which is in itself a cognitive ability: many mammals modulate their behaviour toward others and decide who to cooperate with based on their past interactions and

shared history with other individuals, whereas coalition formation in insect societies relies largely on innate or cue-based recognition. As most swarm robotic models follow the insect paradigm, they lack robot-specific memory and omit interaction histories. However, incorporating such histories appears crucial for modelling experience-based in-group bias, as well as mammal-like social dynamics more broadly. To this end, we introduce two novel cognitive and behavioural mechanisms to existing models: *robot individuation*, in which each robot maintains a partner-specific memory of previous encounters, and *dynamic sociality*, whereby robots' tendency to interact with others is based on their particular history rather than on a fixed sociality parameter.

In this article, we study three intertwined dynamics: foraging, communication, and adaptive sociality. The first two are well known in swarm robotics research, with foraging typically used as the classic ecological testbed, and the Minimal Naming Game (MNG) typically used for studying language emergence [1]. The third component, as far as we are aware, is novel to swarm robotics, with robots operationalized as unique individuals who also keep track of their past interactions with other unique individuals, and modulate their behaviour toward them accordingly. Within this integrated framework, we examine whether introducing these more mammal-like social and cognitive layers leads to the spontaneous emergence of in-group bias from repeated local interactions, most notably without predefined group structure besides spatial proximity (or nest origin). Importantly, although nest origin indirectly affects interaction frequencies, robots do not perceive or encode nest membership, and have no way of knowing to which nest other robots belong. We further investigate how emergent social asymmetries are modulated by robots' memory, their baseline sociality (a so-called *innate* tendency to cooperate, a level of cooperation that is typically high for all agents in swarm robotics), as well as other social and ecological conditions previously proposed to impact social dynamics (group size and the spatial distribution of resources).

Our goal is to identify the minimal cognitive and ecological requirements under which such fundamental social dynamics can emerge in embodied collective systems, so that in the future we could leverage these systems to study complex behaviours that are typical of primates and humans.

2 The Model

Robots are distributed across two nests (marked on the floor as grey disks) and engage in a foraging task: each robot explores the environment, aiming to retrieve items from scattered resources.

2.1 Foraging

We implemented a probabilistic finite state machine (PFSM), shown in Fig 1*a*, which guides robots' behaviour. This PFSM is composed of three states:

- **Exploring**: robots explore their surroundings by following a random walk.

- **Going to nest**: robots move toward the position of their nest.
- **Going to resource**: robots move toward the position of a known resource.

Robots can learn about new resources by reaching an item (discovery)[3]. When robots find an item, they remember its location as a resource area, later targeted in the *Going to resource* state. Robots return to the nest when they have found an item, are rejected during an interaction with another robot, or with probability $P_{return} = 0.0001$. This latter scenario ensures that exploration is centered around the nests, and allows experiments in an open environment. Once robots reach the nest, they immediately go back to either exploring or moving toward a known resource. If robots encounter an item on their path toward a known resource, they pick it up, bring it back to the nest, and update the location of this resource in memory. If a robot manages to reach a known resource position without interruption and without finding a food item, it forgets the resource position and switches back to the exploring state. Notably, this PFSM requires robots to remember landmark positions. This can be achieved through odometry and social feedback, as robots correct their position estimates by exchanging them with nearby peers [8, 17].

2.2 Minimal Naming Game

In parallel to foraging, robots can play a Minimal Naming Game (MNG) [13] with local neighbours (Fig 1*b*). The MNG is a category of language games modelling the cultural evolution of language in populations of agents [29]. In this context, language is operationalized as an evolving lexicon (a list of words, represented here as an 8-bit unsigned vector), specific to each robot. The MNG ensures that the lexicon of all robots in a *well mixed* swarm will converge on a single word. Note that words do not refer to any physical object, but are rather an abstract representation of robots' linguistic variants. When playing this language game, a robot takes one of two roles: sender (with a small probability $P_{send} = 0.01$) or receiver (decided based on partner-specific social history, as detailed in Section 2.3). The sender starts the game by selecting a random word from its lexicon (if the lexicon is empty, it creates a word first) and then broadcasts it within a given range. If the interaction is accepted by the receiver, they check the word against the content of their lexicon. If the word already appears in their lexicon, the game is considered a success, and the receiver keeps only that word in their lexicon, deleting other stored variants. If the word is not already in the lexicon, the game is considered a failure, and the receiver appends the word to their lexicon. Finally, the receiver signals whether the game was a success or a

[3] Since our objective was to first understand how primate-like social dynamics interact with linguistic exchange, we intentionally omitted recruitment and other cooperative foraging mechanisms, as these would create an even stronger group-level structure by design and would mechanically reinforce any in-group bias for reasons unrelated to the social layer considered in this work. As such, current foraging primarily serves to structure the swarm's spatial distribution and interaction patterns in a realistic yet controlled manner.

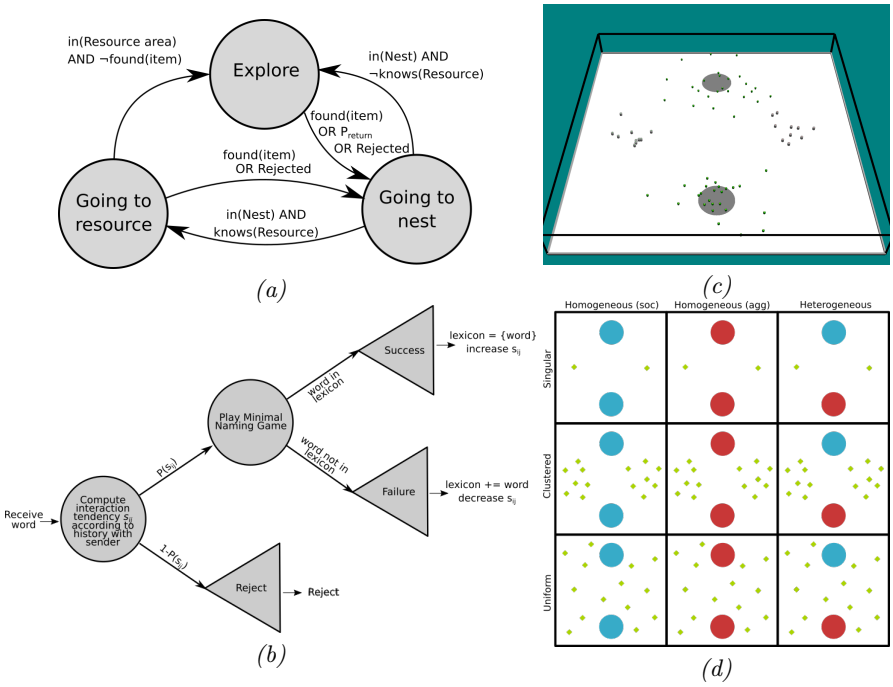

Fig. 1. The behaviour automaton *(a)* guides robots foraging strategy. In any state, a robot can become a *sender* in a Minimal Naming Game (MNG). A nearby robot can then become a *receiver* in this interaction, and decides on its reaction according to the interaction decision-tree *(b)*. The robots are embodied in a simulated world *(c)* with a 2D physics engine, populated by two nests and randomly distributed food items. We observe social and linguistic outcomes in 9 environments *(d)*, with nests inhabited by either aggressive (red) or social (blue) robots, and distributions of items (green).

failure to the sender, who in the case of success, also deletes all other words in their lexicon, keeping only the selected word. In our study, the MNG provides a structured interaction allowing sociality to shape language dynamics.

2.3 Partner-specific sociality

The novelty of our approach is in the introduction of two novel cognitive features. The first is robot individuation, whereby robots have a partner-specific memory and keep track of the outcomes of past interactions with other robots. Crucially, robots are not able to tell whether or not another robot comes from the same nest. Let $mng_{ij}^{k} \in \{0, 1\}$ denote the outcome of the k-th validated MNG between robots i and j (1 for a success, 0 for a failure). We define K_{ij} as the total number of interactions between robots i and j, and let robot i maintain a fixed-size memory window $\mathcal{B}_{ij}^{K_{ij}}$ of the last w validated outcomes with robot j:

$$\mathcal{B}_{ij}^{K_{ij}} = \left(mng_{ij}^{K_{ij}}, mng_{ij}^{K_{ij}-1}, \ldots, mng_{ij}^{K_{ij}-w+1}\right) \tag{1}$$

The second feature is flexible (or evolving) sociality: robots' tendency to interact with others (i.e., their willingness to accept the role of receiver in an incoming MNG) is based on the degree of sociality toward the specific sender robot. This robot-specific sociality value is based on prior experience: successful communication between two robots increases their sociality toward each other (and therefore their willingness to communicate again), while a failed communication decreases their sociality toward each other. More specifically, we compute sociality (or the probability of interacting) $s_{ij}(K_{ij}, t)$ as a weighted moving average of the number of successful past interactions between robots i and j:

$$s_{ij}(K_{ij}, t) = \frac{1}{w} \sum_{k=K_{ij}-w+1}^{K_{ij}} \left(d_{ij}^k \times mng_{ij}^k + (1 - d_{ij}^k) \times s_i^0 \right), \quad \text{with } d_{ij}^k = e^{-\lambda\,(t - t_{ij}^k)}, \quad mng_{ij}^k \in \{0, 1\} \tag{2}$$

where s_i^0 is the baseline sociality of robot i, t_{ij}^k the time of the k-th interaction between both robots, w the memory window size, and λ the memory decay parameter. The latter controls how quickly the influence of past interactions fades, gradually pulling $s_{ij}(K_{ij}, t)$ back toward the baseline value s_i^0. To stabilize early dynamics, we introduce a prior consisting of a pre-filled backlog of interactions. We make them infinitely old, hence neutral toward the baseline sociality s_i^0 because of decay.

3 Experimental setup

We set up a swarm of robots allocated to two nests, in a 10m × 10m enclosed arena. The arena was large enough so that robots would (almost) never reach its limits, making it akin to an open environment. At the center of the arena, we placed two nests (above and below the midpoint) and two resources (left and right), creating a symmetrical layout (Fig 1*c*). Both nests are 1.2 meters in diameter and 3 meters apart. Resources are equidistant from each nest and 2 meters apart from one another. We examine the emergence, strength, and stability of in-group bias across ecological, cognitive, and social conditions:

- **Ecological Parameters**: items are distributed according to a Gaussian kernel centered on each resource, with standard deviations $\sigma \in \{0.1, 1, 10\}$, corresponding to *singular*, *clustered*, and *uniform* distributions. Each resource holds up to 25 items; when below capacity, new items appear automatically following their respective Gaussian kernel.
- **Cognitive Parameters**: the evolution of robots' sociality depends on two parameters: how well robots remember their past interactions, controlled by the memory–decay rate $\lambda \in \{0.001, 0.0001, 0.00001\}$, and the size of their bounded memory window, $w \in \{10, 30, 60\}$.
- **Social Parameters**: each robot i has a baseline sociality value $s_i^0 \in \{0.1, 0.9\}$, corresponding respectively to initially *aggressive* or *social* robots. We test both homogeneous (aggressive–aggressive, social–social) and heterogeneous

(aggressive–social) nest configurations, and vary the number of robots per nest with population size $N \in \{12, 25, 50\}$.

For each experimental setup, we performed 50 runs, carried out with ARGoS3 [22], a physically realistic simulator made for swarm robotics experimentation, with a simulation rate of 10 ticks per second. We simulated e-puck robots [18, 5], a robot that uses a range-and-bearing system to communicate locally [7].

4 Results

4.1 Ecological conditions

We first evaluate whether the emergence of social asymmetries is impacted by ecology. Although resource topology could in principle shape interaction patterns, our results reveal a surprisingly stable outcome across conditions. Indeed, we find that in-group bias consistently emerges in all ecological configurations (*singular*, *clustered* and *uniform* resource spread). Similarly, none of the resource distributions produced substantial differences in social or linguistic outcomes (see supplementary material). This robustness may stem from the individual nature of the foraging task; without recruitment or coordinated search (which we plan to reintroduce in future work), resource topology minimally affects robots' movement and encounter rates. Given this consistency, we adopt a uniform resource distribution for all subsequent experiments.

4.2 Cognitive conditions

Next, we examine how sociality varies as a function of the rate of memory–decay and the size of the memory window. In the following experiments, we have $N = 25$ robots per nest, focusing on homogeneous populations where all robots are initialized with the same baseline sociality (Fig 2).

Memory decay λ and memory window size w shape social dynamics similarly. Smaller memory windows and slower decay make robots update their sociality levels quicker: a lower w gives disproportionate weight to each new interaction, and a slower decay preserves that influence longer. Conversely, large memory windows introduce substantial inertia, and a high memory decay prevents robots from diverging from their baseline sociality. For social robots, sociality slightly increases toward both in-group and out-group members, resulting in minimal preference given their already near-ceiling sociality. On the other hand, aggressive robots, starting from a low sociality baseline, develop a strong and persistent in-group bias, with sociality toward out-group remaining almost null, while sociality toward in-group reaches levels that are nearly as high as social robots.

We see that memory decay has little effect at the beginning of the simulation when past interactions have not yet accumulated. At this stage, each interaction contributes roughly $1/w$ to the sociality estimate. Therefore, smaller memory window sizes produce much stronger early fluctuations, accelerating divergence between in-group and out-group. A higher λ gives an upper bound to the sociality

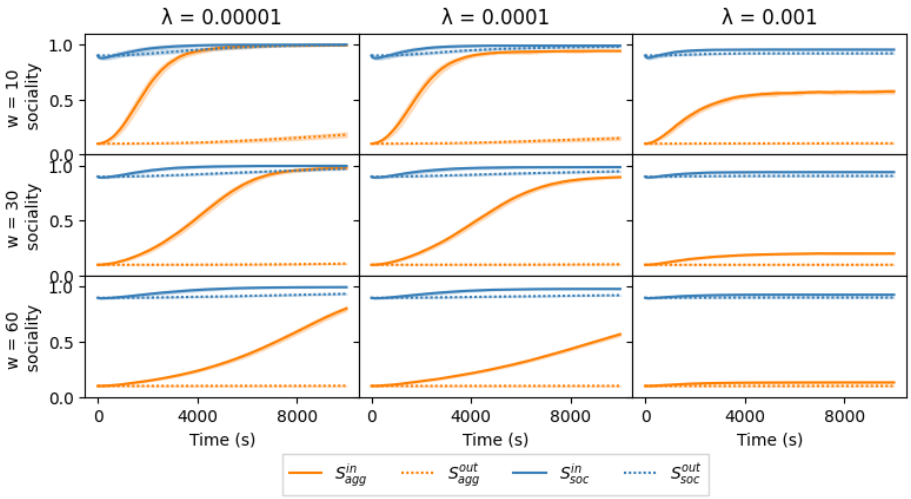

Fig. 2. Evolution of the overall sociality over time in a homogeneous configuration (both sub-swarms have the same baseline sociality s_i^0) for different values of decay λ (columns), memory window size w (rows) and baseline sociality (orange for aggressive robots, blue for social robots). Curves for both sociality conditions (aggressive and social robots) are plotted together in each panel. Solid lines correspond to in-group interactions, and dashed lines to out-group interactions. Overall, initially highly social robots remain equally willing to interact with both in- and out-groups, while robots with initial low sociality gradually become more social toward in-group robots (solid orange) but maintain low sociality toward out-group robots (dashed orange).

curves, reinforced with a lower w. Intuitively, this happens because after reaching a certain sociality, the rate at which new successes enter the window becomes roughly constant, and counterbalances the continuous decay of older entries, creating a balance between gain and loss that naturally bounds the curve.

Next, we analyse MNG activity, social interaction patterns, and vocabulary convergence, with $\lambda = 0.0001$, $w = 30$ and $N = 25$ (Fig 3). Social homogeneous swarms show the baseline pattern expected for highly sociable populations (default swarm robotics behaviour). Cross-group interactions gradually increase as robots explore the arena and are not rejected by others (neither in-group nor out-group). As a result, the rate of MNGs played between the two social groups increases. Over time, both swarms grow increasingly similar. Aggressive robots, by contrast, begin with few interactions overall, regardless of group membership. Early rejections frequently interrupt exploration and send robots back to their nest, leading to more in-group interactions. This contributed to a strong in-group bias in both interaction frequency and MNG rate: robots play significantly more MNGs with in-groups and almost none with out-groups. Social robots achieve quick lexical convergence thanks to frequent MNGs played even with out-group robots, whereas aggressive robots interact far less and thus converge more slowly.

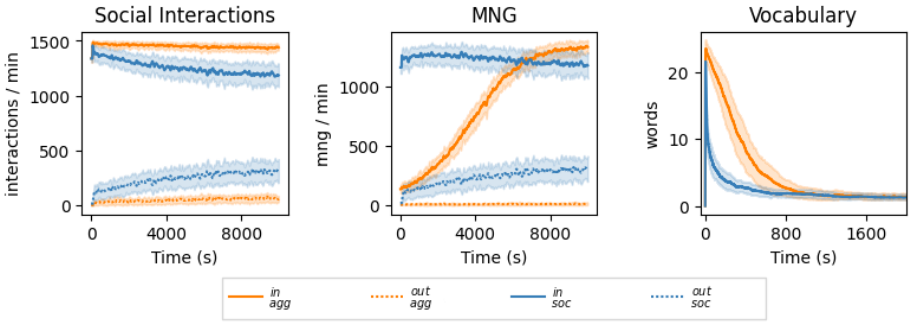

Fig. 3. Evolution, per swarm, of, left to right, the overall rate of social interactions initiated over time, the rate of naming games played over time, and the number of words present in the swarm's lexicon for different baseline sociality values (colors). Curves for both baseline sociality conditions (aggressive and social) are plotted together in each panel. Please note the different time scale used for the vocabulary panel.

4.3 Social conditions

Group size Next, we examine how group size affects swarm social dynamics. In this section, we keep $\lambda = 0.0001$ and $w = 30$ (Fig 4). Although more robots increase the number of potential interactions, each robot must interact with a larger set of partners. In practice, the rise in interactions does not counterbalance the difficulty of robots reaching one another exhaustively, resulting in slower social dynamics. This is partly driven by physical congestion: with more robots in the arena, collisions and impeded navigation become more frequent, making it harder to reliably reach one another and leaving the swarm less *well mixed*.

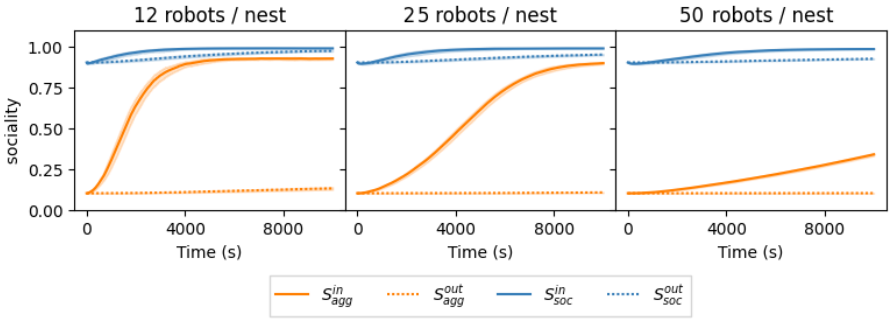

Fig. 4. Evolution of overall swarm sociality in a homogeneous configuration for different number of robots per nest (columns) and baseline sociality (colors). Curves for both sociality conditions (aggressive and social robots) are plotted together in each panel.

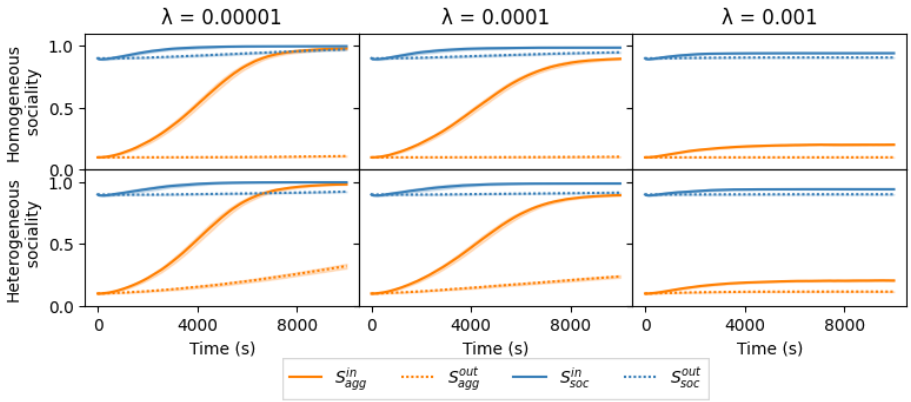

Fig. 5. Evolution of sociality over time in homogeneous configuration (top, both sub-swarms have the same baseline sociality) and heterogeneous configuration (bottom, each sub-swarm has a different baseline sociality value) for different values of decay λ (columns) and memory window size $w = 30$. In heterogeneous contexts, the two colors of curve correspond to each sub-swarm (social or aggressive) of the same experience.

Heterogeneous scenario Finally, we examine the impact of different baseline sociality. In the following experiments, we keep $\lambda = 0.0001$, $w = 30$ and $N = 25$, focusing on heterogeneous populations where one nest is initialized with a high baseline sociality and the other with a low sociality (see Fig 5). Introducing heterogeneity does not substantially alter within-group interactions. In-group patterns mirror those in homogeneous settings. Social robots show reduced sociality toward the out-group (aggressive robots), as they are rejected more often. On the other hand, aggressive robots are influenced by social out-group robots: their sociality increases compared to purely out-group aggressive robots, though an in-group bias is still present. High decay rates (or small memory windows) allow sociality to increase so quickly that aggressive robots become more open to out-group interactions, which gradually reduces the in-group bias.

5 Conclusion

Our paper set out to model the spontaneous emergence of a well-documented social asymmetry, namely the in-group bias, which governs animals' social dynamics in the wild. Rather than being innate, in mammals and humans this social bias is acquired over time, and evolves according to prior history and flexible group membership representations. To this end, we created modified swarms with novel cognitive abilities like memory and individual recognition, mimicking some of the complex social dynamics that guide mammalian behaviour. We then tested the minimal conditions that give rise to this bias, and how its strength and stability are affected by various cognitive, social, and ecological variables.

Our results show that an in-group bias can spontaneously emerge in a swarm of embodied agents. With the introduction of mammal-like cognitive features, robots gradually differentiated between individuals they encountered frequently and those they encountered rarely. While highly social robots maintain broad cooperation and show only mild asymmetries (as expected from classic simulations where all robots have a maximal tendency to cooperate), we find that initially aggressive robots develop a strong in-group bias, where robots from their own nest are favoured (i.e., high sociality values), while robots from the other nest are rejected (i.e., low sociality values). Importantly, these effects arise purely from interaction frequencies shaped by spatial structure, and not from any a priori encoded group membership. In other words, cumulative patterns of interaction lead robots to develop strong group membership associations, effectively forming sub-swarms that tend to communicate mostly among themselves.

The emergence of in-group bias appears robust across ecologies, with different resource distributions having little influence on the model outcomes. In contrast, cognitive parameters such as memory size and decay rate strongly affected how quickly social patterns emerged: larger, more stable memories slow down social dynamics for both in-group and out-group, potentially delaying the development of in-group bias while making it more persistent once established. Group size played a role as well, with larger swarms displaying slower and more variable dynamics due to congestion and reduced mixing. In heterogeneous scenarios, social robots could partly pull aggressive robots toward higher sociality, eventually eliminating group bias over time with small memory size and low decay. When memory is robust, a reliable in-group bias emerges over time.

Overall, our work demonstrates that simple social-cognitive extensions allow swarms of robots to reproduce key aspects of mammal-like social dynamics, offering a physically grounded framework to study emergent social structure and cultural divergence in more complex species.

5.1 Future Work

Our broader goal here is to create more primate-like swarms that can help model human behaviour. One natural extension is to reintroduce cooperative foraging, fully integrated with social and linguistic dynamics, rather than as a separate process. Another direction is to explore the relative time scales of foraging, linguistic convergence, and social adaptation, which our results suggest strongly shape group structure. In particular, we plan to study patterns of linguistic convergence (same or different words across sub-swarms) depending on the sociality of the robots. Finally, moving beyond fixed nests and toward naturalistically emergent territories would bring the model closer to mammalian spatial organisation. This would diversify movement patterns beyond simple nest–resource exploitation loops. This increased spatial complexity could, in turn, create favourable conditions for richer, potentially compositional communication systems.

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